

SIC Smart Bank System

SIC 7

Coding and Programming

G 9

Agenda

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Agenda

- | Introduction & Problem Statement
- | **Solution Overview & User Flow**
- | Core User Features
- | Transaction Logic
- | ATM & Admin Analysis
- | Python Concepts Used
- | Results & Demonstration
- | Q&A

INTRODUCTION

A Python-based console banking system designed to manage users, wallets, transactions, ATMs, and admin analysis.

PROBLEM STATEMENT

- No centralized user management
 - No transaction history tracking
- No admin analytics or user segmentation
 - No ATM status monitoring

TARGET USERS

- End-users who need basic banking operations
- Administrators who need user and transaction insights

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SOLUTION

A Python-based console application that provides separate paths for users and administrators.

USER FLOW

Register → Login → Wallet Menu

ADMIN FLOW

Admin Login → Admin Report

KEY FEATURES

- User registration and login
- Wallet and transaction operations
 - ATM status monitoring
- Admin reports and analysis

Core User Features
Account Management
User Registration
Secure Login
Unique User ID
Failed Login Attempts
Wallet Operations
Deposit Money
Withdraw Money
Transfer Funds
Currency Conversion
Profile Management
Update Personal Information
Phone & City
Password Management
Emergency Contact
Screenshot: Wallet Menu

```
def wallet_menu(user):  
    1 usage  
    print("Invalid receiver.")  
    continue  
  
    try:  
        amount, currency = input("Enter amount and currency: ")  
        amount = float(amount)  
    except ValueError:  
        print("Invalid input.")  
        continue  
  
    currency = currency.upper()  
    if amount <= 0 or currency not in rates:
```

Run console output:

```
Successful. Balance: 2500.0 EGP  
Welcome Omar Hassan  
[0] Deposit  
[1] Withdraw  
[2] Transfer  
[3] Transaction History  
[4] Reports  
[5] ATM status  
[6] Update personal info  
[7] Exit  
> 3  
First: {'type': 'Deposit', 'amount': 2500.0, 'currency': 'EGP', 'value_egg': 2500.0}  
Last: {'type': 'Deposit', 'amount': 2500.0, 'currency': 'EGP', 'value_egg': 2500.0}  
{'type': 'Deposit', 'amount': 2500.0, 'currency': 'EGP', 'value_egg': 2500.0}
```

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How It Works: Transaction Logic

Currency Conversion

EGP = 1

USD = 30 EGP

SAR = 9 EGP

Deposit

Amount × Currency Rate → EGP Balance

Withdrawal

Check Balance → Deduct Amount → Record Transaction

Transfer

Find the receiver

Check available balance

Deduct from sender

Add to receiver

Record the transaction

Transaction History

First transaction

Last transaction

Up to the last 5 transactions

Screenshot: Deposit / Withdrawal / Transfer screen

The screenshot displays a PyCharm IDE interface. The left sidebar shows a project named 'PythonProject' with a file explorer containing various Python files like '9.py', 'a.py', 'Analyzer program.py', 'Café Menu Manager.py', 'Course Registration Checker.py', 'd.py', 'data.json', 'f.py', 'Favorite_Things.py', 'guess_number.py', and 'h.py'. The main editor window shows a Python function 'def wallet_menu(user):' with several lines of code, including input prompts, error handling with 'except ValueError', and currency conversion logic. The bottom 'Run' console shows the program's output: 'Welcome Omar Hassan', a menu of options [0] Deposit through [7] Exit, and a successful transaction confirmation: 'Enter amount and currency: 2500 EGP' followed by 'Successful. Balance: 2500.0 EGP'.

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How It Works: ATM & Admin Analysis

ATM Status System

1 = Available

0 = Out of Service

Display ATM status using a 4×4 matrix

Count available and unavailable ATMs

Update individual ATM status

Admin Analysis

Display User IDs and Names

Detect duplicate emails and phone numbers

Count transaction frequency

Analyze and segment users

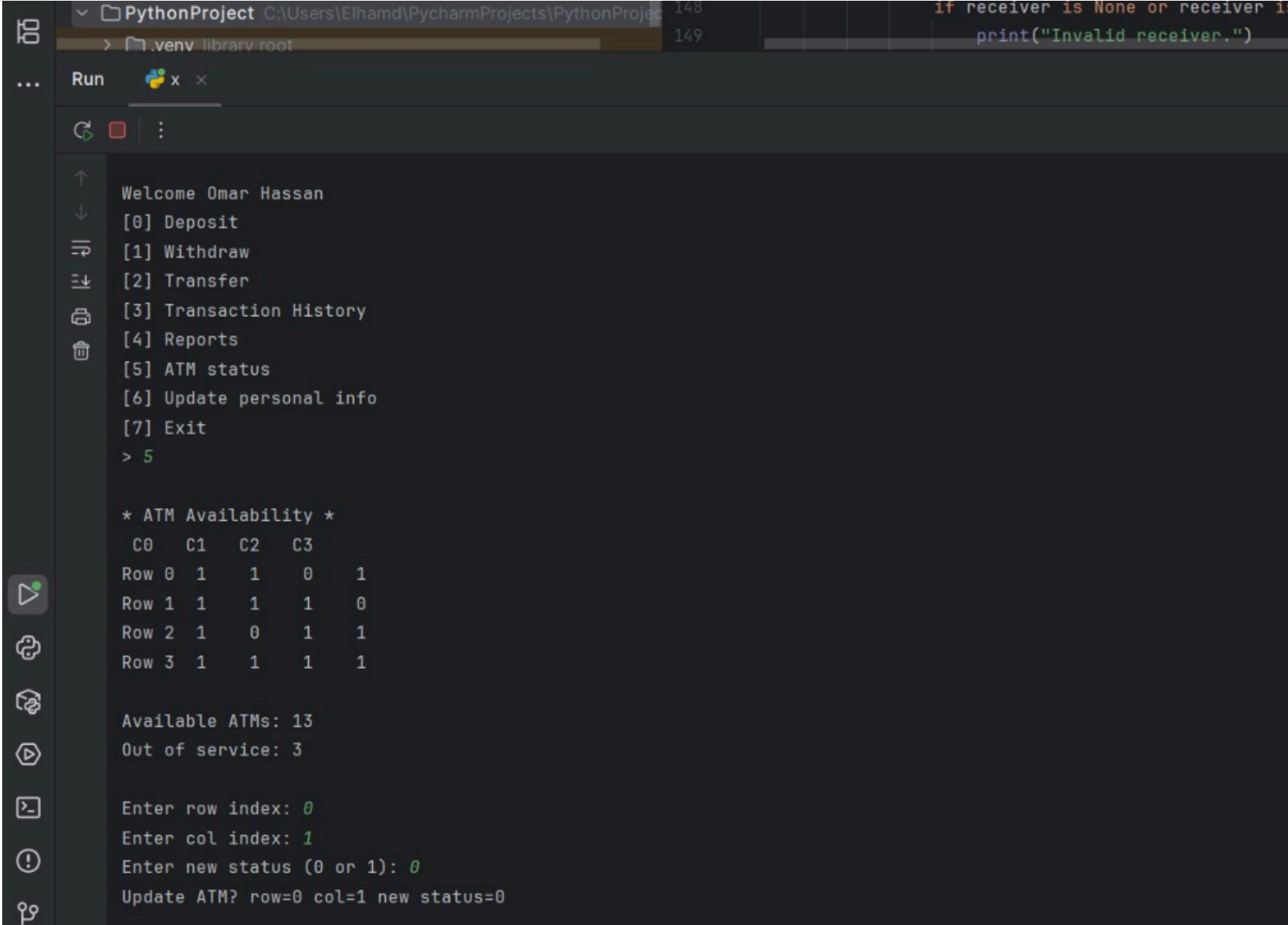
Set Operations

Union ($\cup$) → VIP + Active Users

Intersection ($\cap$) → VIP $\cap$ Active Users

Difference ($-$) → Users in only one group

Screenshot: 4×4 ATM Matrix



```
PythonProject C:\Users\Elhamd\PycharmProjects\PythonProject 148 if receiver is None or receiver is
> .venv library root 149 print("Invalid receiver.")

Run x x

Welcome Omar Hassan
[0] Deposit
[1] Withdraw
[2] Transfer
[3] Transaction History
[4] Reports
[5] ATM status
[6] Update personal info
[7] Exit
> 5

* ATM Availability *
C0 C1 C2 C3
Row 0 1 1 0 1
Row 1 1 1 1 0
Row 2 1 0 1 1
Row 3 1 1 1 1

Available ATMs: 13
Out of service: 3

Enter row index: 0
Enter col index: 1
Enter new status (0 or 1): 0
Update ATM? row=0 col=1 new status=0
```

Python Concepts Used

Data Structures

- Lists → Store users and transactions
- Dictionaries → Store user and transaction data
- Sets → Detect duplicates and compare users
- Tuples → Store fixed collections of data

Control Flow

- if / elif / else
- for loops
- while loops
- try / except

Key Functions & Methods

- append() → Add data
- pop() → Remove data
- count() → Count elements
- update() → Update dictionary data
- zip() → Combine related data
- enumerate() → Track ATM positions

Add a little bit of body text

Results & Demonstration

System Demonstration

Registration & Login

Create an account and securely access the system.

Banking Operations

Deposit, withdraw, and transfer money between users.

ATM Management

View and update ATM availability using a 4×4 matrix.

Admin Analysis

Generate reports and analyze user and transaction data.

```

Welcome Omar Hassan
[0] Deposit
[1] Withdraw
[2] Transfer
[3] Transaction History
[4] Reports
[5] ATM status
[6] Update personal info
[7] Exit
> 4
Balance: 2500.0 EGP
Deposits: 2500.0 Withdrawals: 0
Sent: 0 Received: 0
    
```

```

=== SIC SMART BANK SYSTEM ===
If you already have an account, enter login
If you do not have an account, enter register
To close the system, enter exit
> login

--- Login ---
Please enter your ID: 4
Please enter your password: Omar@1234
Login successful! Welcome back, Omar Hassan

Welcome Omar Hassan
    
```

```

C:\Users\Elhamd\PycharmProjects\PythonProject\.venv\Scripts\python.exe C:\Users\Elhamd\Py

=== SIC SMART BANK SYSTEM ===
If you already have an account, enter login
If you do not have an account, enter register
To close the system, enter exit
> register

--- Register ---
Please enter your name: Omar Hassan
Please enter your password: Omar@1234
Please enter your phone number: 01555555555
Please enter your email: omar.test@gmail.com
Please enter your gender: male
Please enter your age: 22
Please enter your city: Port Said
Please enter your account type: regular
Sign up successful. Your ID is 4
    
```

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Conclusion & Q&A

Key Takeaway

A functional banking system that applies fundamental Python concepts to solve a real-world problem.

What We Achieved

User management and authentication

Core banking operations

ATM status management

Admin analytics and reporting

Practical use of Python data structures and logic

Future Improvements

Database integration

Graphical User Interface (GUI)

Live exchange rates using APIs

Enhanced security

Thank You

Questions?